POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 6023C | Semester VI | Theory | 4 Credits

Electric Vehicles

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6023C as Electric Vehicles in Semester VI for Mechanical Engineering. The official SITTTR detailed course page was unavailable. With the user's approved public-source approach, this handbook is transparently based on the NPTEL IIT Madras course Fundamentals of Electric Vehicles: Technology & Economics and the U.S. Department of Energy Alternative Fuels Data Center overview of all-electric vehicle components. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. High-voltage systems, traction batteries, charging equipment, damaged-vehicle handling, thermal events and emergency response require current manufacturer procedures, applicable regulations, insulated tools/PPE and appropriately trained/authorised personnel.

Verified source note: Verified sources support EV architecture, traction forces/drive cycles, drivetrain design, battery parameters, SoC/SoH, pack mechanical/thermal/electrical design, BMS, motor/controller concepts, charging/swapping, system components and infrastructure. The four modules form a study sequence rather than an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Electric Vehicles introduces the vehicle-level integration of traction energy storage, power electronics, electric machines, drivetrains, thermal systems, charging and control. The course develops an engineering view of vehicle performance, battery management, infrastructure, quality and safety without substituting for authorised high-voltage service training.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□ □□□□□□□□.

Prerequisite foundation

Basic electrical engineering, mechanics/vehicle fundamentals, thermodynamics/heat transfer awareness, measurement concepts and strict electrical/battery safety discipline.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain EV types, system architecture, vehicle-resistance forces and drive-cycle energy concepts.
2. Explain battery fundamentals, pack design, state estimation, thermal management and BMS roles.
3. Explain traction motors, controllers, power-electronics interfaces and drivetrain integration.
4. Explain charging, swapping, infrastructure, sustainability and high-voltage safety responsibilities.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□□ exam-□ □□□□□□ □□□□□□□□ □□□□□□□ □□□□□□ □□□□□□□□□. Define, explain, apply □□□□□□ action words □□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain EV architectures, subsystems, resistance forces and vehicle-performance concepts.	Understand
CO2	Explain battery-cell/pack fundamentals, SoC/SoH, BMS functions and thermal/electrical design considerations.	Understand
CO3	Explain traction motors, controllers, power-electronics interfaces and drivetrain energy flow.	Understand
CO4	Explain charging/swapping, infrastructure, sustainability and high-voltage safety responsibilities.	Understand

Official CO-PO articulation matrix							
CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7–PO11**
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	EV Architecture and Vehicle Performance	EV types and architecture; key subsystems; traction/auxiliary energy paths; rolling, aerodynamic and grade resistance; power, torque and drive-cycle energy concepts.	Approx. 14 hours	CO1	Module 1
2	Batteries, Packs, Thermal Design and BMS	Cell and pack fundamentals; energy/power trade-off; battery parameters; SoC/SoH and self-discharge awareness; mechanical, electrical and thermal pack design; BMS functions.	Approx. 17 hours	CO2	Module 2
3	Motors, Controllers and Drivetrain Integration	Electric-machine roles; torque, speed and back-EMF concepts; inverter/controller, DC/DC and onboard charger roles; regenerative braking; drivetrain and thermal integration.	Approx. 15 hours	CO3	Module 3
4	Charging, Infrastructure, Sustainability and Safety	AC/DC charging, charging rate and infrastructure; swapping concepts; system communication/standardisation awareness; lifecycle/sustainability; inspection, incident awareness and safety boundaries.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

EV Architecture and Vehicle Performance

EV types and architecture; key subsystems; traction/auxiliary energy paths; rolling, aerodynamic and grade resistance; power, torque and drive-cycle energy concepts.

CO1 · Approx. 14 hours

Batteries, Packs, Thermal Design and BMS

Cell and pack fundamentals; energy/power trade-off; battery parameters; SoC/SoH and self-discharge awareness; mechanical, electrical and thermal pack design; BMS functions.

CO2 · Approx. 17 hours

Motors, Controllers and Drivetrain Integration

Electric-machine roles; torque, speed and back-EMF concepts; inverter/controller, DC/DC and onboard charger roles; regenerative braking; drivetrain and thermal integration.

CO3 · Approx. 15 hours

Charging, Infrastructure, Sustainability and Safety

AC/DC charging, charging rate and infrastructure; swapping concepts; system communication/standardisation awareness; lifecycle/sustainability; inspection, incident awareness and safety boundaries.

CO4 · Approx. 14 hours

MODULE 1

EV Architecture and Vehicle Performance

EV types and architecture; key subsystems; traction/auxiliary energy paths; rolling, aerodynamic and grade resistance; power, torque and drive-cycle energy concepts.

Approx. 14 hours | **CO1** | **Understand · Apply**

Learning goals

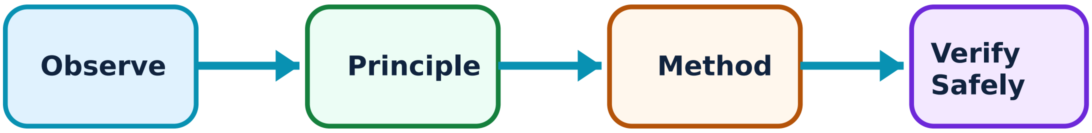
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for EV Architecture and Vehicle Performance: identify the system, state its principle, follow the correct method and verify the result safely.

1. EV configurations and energy path−

Battery-electric vehicles use a traction battery to supply an electric drive system. Core elements include charge port, traction and auxiliary batteries, power electronics, motor, thermal system, drivetrain/transmission and control interfaces. Hybrid and fuel-cell vehicles use different energy architectures.

Study and application method

Draw a high-level energy-flow diagram from charging source to battery, power electronics, motor and wheels, plus the auxiliary and thermal branches. State the role of each block.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Draw a high-level energy-flow diagram from charging source to battery, power electronics, motor and wheels, plus the auxiliary and thermal branches. State the role of each block.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: A conceptual diagram is not a service map. Do not touch, probe or disconnect high-voltage components without approved training, equipment and isolation verification.

2. Forces, torque, power and drive cycles−

Vehicle energy demand is influenced by rolling resistance, aerodynamic drag, road grade, mass, acceleration and drive-cycle behaviour. Motor torque and speed requirements are matched to these loads through the drivetrain and controls.

Study and application method

For a supplied scenario, list the dominant resistances at low/high speed, identify the data needed to estimate wheel force/power and explain how a drive cycle changes energy use.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

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Common mistake / precaution: Simplified vehicle calculations omit control limits, thermal effects, regenerative braking limits, tyres, weather and accessory loads; state all assumptions.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Batteries, Packs, Thermal Design and BMS

Cell and pack fundamentals; energy/power trade-off; battery parameters; SoC/SoH and self-discharge awareness; mechanical, electrical and thermal pack design; BMS functions.

Approx. 17 hours | CO2 | Understand · Apply

Learning goals

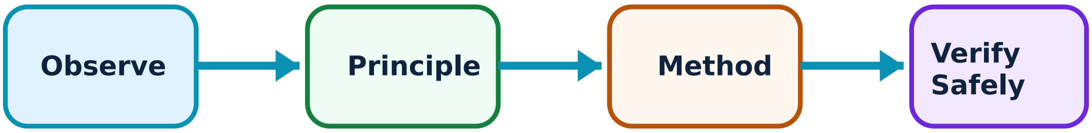
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Batteries, Packs, Thermal Design and BMS: identify the system, state its principle, follow the correct method and verify the result safely.

1. Battery parameters and pack architecture−

A traction pack connects cells/modules with electrical protection, sensing, structural support, cooling/heating and high-voltage interfaces. Capacity, voltage, energy, power, internal resistance, temperature and ageing influence usable performance and range.

Study and application method

Prepare a battery-system table that separates cell, module and pack levels; record the parameter, unit, measurement condition, system effect and reason it must be monitored.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Prepare a battery-system table that separates cell, module and pack levels; record the parameter, unit, measurement condition, system effect and reason it must be monitored.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Battery packs can retain hazardous energy even when a vehicle is off. Never open, crush, short, heat or test cells/packs outside a controlled authorised facility.

2. SoC, SoH, thermal management and BMS−

State of charge estimates available energy; state of health indicates ageing/performance condition. A BMS monitors voltage, current and temperature, supports balancing/protection and communicates with vehicle/charging systems. Thermal design helps keep components within intended operating limits.

Study and application method

Trace a BMS information flow from cell sensors to decision logic, contactor/protection action, driver/charger communication and data record. Identify one uncertainty in each measurement.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Trace a BMS information flow from cell sensors to decision logic, contactor/protection action, driver/charger communication and data record. Identify one uncertainty in each measurement.

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Common mistake / precaution: BMS functions and fault responses are safety-critical and manufacturer-specific. Do not bypass interlocks or treat a dashboard indication as proof that a pack is safe.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Motors, Controllers and Drivetrain Integration

Electric-machine roles; torque, speed and back-EMF concepts; inverter/controller, DC/DC and onboard charger roles; regenerative braking; drivetrain and thermal integration.

Approx. 15 hours | CO3 | Understand · Apply

Learning goals

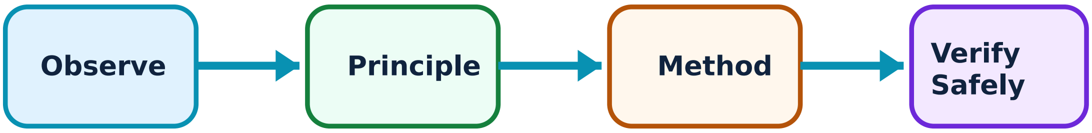
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Motors, Controllers and Drivetrain Integration: identify the system, state its principle, follow the correct method and verify the result safely.

1. Traction motors and control−

An electric traction motor converts electrical energy to wheel-driving mechanical energy. Controllers/inverters regulate power and torque by managing current/voltage, while motor characteristics such as torque-speed behaviour and back EMF influence system design.

Study and application method

Use an energy-flow sketch to distinguish battery DC, inverter/controller function, motor mechanical output and regenerative flow. Explain why operating region and cooling matter.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use an energy-flow sketch to distinguish battery DC, inverter/controller function, motor mechanical output and regenerative flow. Explain why operating region and cooling matter.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Power electronics can contain high voltage, stored energy and sensitive components. Use only approved measurement methods and discharge/isolation procedures.

2. Power-electronics and vehicle integration−

Power electronics may include traction inverter, DC/DC converter and onboard charger. They coordinate traction, auxiliary supply, charging and protection while interacting with thermal systems, vehicle controls and mechanical drivetrain components.

Study and application method

Map each component’s input, output, function, control signal and cooling need. Compare its role during driving, regenerative braking and AC charging.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Map each component's input, output, function, control signal and cooling need. Compare its role during driving, regenerative braking and AC charging.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Interoperability and diagnostics depend on vehicle-specific architecture and software. Never substitute components or alter settings without authorised engineering approval.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Charging, Infrastructure, Sustainability and Safety

AC/DC charging, charging rate and infrastructure; swapping concepts; system communication/standardisation awareness; lifecycle/sustainability; inspection, incident awareness and safety boundaries.

Approx. 14 hours | CO4 | Understand · Apply

Learning goals

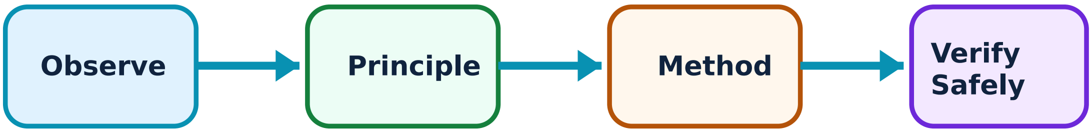
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Charging, Infrastructure, Sustainability and Safety: identify the system, state its principle, follow the correct method and verify the result safely.

1. Charging and swapping systems−

Charging connects the vehicle to external power through EV supply equipment; onboard or external equipment manages conversion and control. Charging rate, connector/system compatibility, grid capacity, thermal limits, user behaviour and site design affect performance. Swapping exchanges standardised battery packs under controlled conditions.

Study and application method

Compare a charging scenario by power source, AC/DC conversion location, vehicle/EVSE communication, expected constraints, site data needed and safety controls—without calculating or instructing live installation.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare a charging scenario by power source, AC/DC conversion location, vehicle/EVSE communication, expected constraints, site data needed and safety controls—without calculating or instructing live installation.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Do not design, install or modify charging equipment from a handbook. Electrical installation and public charging require certified design, inspection, protection and regulatory compliance.

2. Sustainability, lifecycle and safe response−

EV sustainability depends on electricity source, material extraction, manufacturing, use efficiency, battery life, repair, second use and recycling. Safety planning covers normal operation, damaged vehicles, insulation faults, thermal events and emergency coordination under authorised procedures.

Study and application method

Create a lifecycle evidence map: stage, potential benefit/impact, data source, uncertainty, responsible stakeholder and improvement question. Add a clear escalation boundary for any high-voltage/battery abnormality.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Create a lifecycle evidence map: stage, potential benefit/impact, data source, uncertainty, responsible stakeholder and improvement question. Add a clear escalation boundary for any high-voltage/battery abnormality.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: If a battery is damaged, overheating, emitting odour/smoke or involved in a collision, follow the applicable emergency/manufacturer procedure and contact trained responders; do not improvise a repair or containment action.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: EV Architecture and Vehicle Performance

Use the concept flow to revise observation, principle, method and verification.

Module 2: Batteries, Packs, Thermal Design and BMS

Use the concept flow to revise observation, principle, method and verification.

Module 3: Motors, Controllers and Drivetrain Integration

Use the concept flow to revise observation, principle, method and verification.

Module 4: Charging, Infrastructure, Sustainability and Safety

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Draw a BEV subsystem and energy-flow diagram, clearly separating traction, auxiliary, thermal and charging functions.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Analyse a supplied simplified drive-cycle case, listing resistance forces, assumptions and energy-use factors.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Create a cell–module–pack–BMS concept map showing parameters, sensors, decisions and safety boundaries.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Compare AC charging, DC charging and swapping at a conceptual level using technical, infrastructure and safety criteria.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Prepare a lifecycle evidence map for an EV/battery case, distinguishing verified evidence from open questions.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — EV Architecture and Vehicle Performance

Explain EV configurations and energy path and state one correct method, application or precaution.−

Battery-electric vehicles use a traction battery to supply an electric drive system. Core elements include charge port, traction and auxiliary batteries, power electronics, motor, thermal system, drivetrain/transmission and control interfaces. Hybrid and fuel-cell vehicles use different energy architectures.

Answer framework: Draw a high-level energy-flow diagram from charging source to battery, power electronics, motor and wheels, plus the auxiliary and thermal branches. State the role of each block.

Important point: A conceptual diagram is not a service map. Do not touch, probe or disconnect high-voltage components without approved training, equipment and isolation verification.

Explain Forces, torque, power and drive cycles and state one correct method, application or precaution.−

Vehicle energy demand is influenced by rolling resistance, aerodynamic drag, road grade, mass, acceleration and drive-cycle behaviour. Motor torque and speed requirements are matched to these loads through the drivetrain and controls.

Answer framework: For a supplied scenario, list the dominant resistances at low/high speed, identify the data needed to estimate wheel force/power and explain how a drive cycle changes energy use.

Important point: Simplified vehicle calculations omit control limits, thermal effects, regenerative braking limits, tyres, weather and accessory loads; state all assumptions.

Module 2 — Batteries, Packs, Thermal Design and BMS

Explain Battery parameters and pack architecture and state one correct method, application or precaution.−

A traction pack connects cells/modules with electrical protection, sensing, structural support, cooling/heating and high-voltage interfaces. Capacity, voltage, energy, power, internal resistance, temperature and ageing influence usable performance and range.

Answer framework: Prepare a battery-system table that separates cell, module and pack levels; record the parameter, unit, measurement condition, system effect and reason it must be monitored.

Important point: Battery packs can retain hazardous energy even when a vehicle is off. Never open, crush, short, heat or test cells/packs outside a controlled authorised facility.

Explain SoC, SoH, thermal management and BMS and state one correct method, application or precaution.−

State of charge estimates available energy; state of health indicates ageing/performance condition. A BMS monitors voltage, current and temperature, supports balancing/protection and communicates with vehicle/charging systems. Thermal design helps keep components within intended operating limits.

Answer framework: Trace a BMS information flow from cell sensors to decision logic, contactor/protection action, driver/charger communication and data record. Identify one uncertainty in each measurement.

Important point: BMS functions and fault responses are safety-critical and manufacturer-specific. Do not bypass interlocks or treat a dashboard indication as proof that a pack is safe.

Module 3 — Motors, Controllers and Drivetrain Integration

Explain Traction motors and control and state one correct method, application or precaution.−

An electric traction motor converts electrical energy to wheel-driving mechanical energy. Controllers/inverters regulate power and torque by managing current/voltage, while motor characteristics such as torque-speed behaviour and back EMF influence system design.

Answer framework: Use an energy-flow sketch to distinguish battery DC, inverter/controller function, motor mechanical output and regenerative flow. Explain why operating region and cooling matter.

Important point: Power electronics can contain high voltage, stored energy and sensitive components. Use only approved measurement methods and discharge/isolation procedures.

Explain Power-electronics and vehicle integration and state one correct method, application or precaution.−

Power electronics may include traction inverter, DC/DC converter and onboard charger. They coordinate traction, auxiliary supply, charging and protection while interacting with thermal systems, vehicle controls and mechanical drivetrain components.

Answer framework: Map each component’s input, output, function, control signal and cooling need. Compare its role during driving, regenerative braking and AC charging.

Important point: Interoperability and diagnostics depend on vehicle-specific architecture and software. Never substitute components or alter settings without authorised engineering approval.

Module 4 — Charging, Infrastructure, Sustainability and Safety

Explain Charging and swapping systems and state one correct method, application or precaution.−

Charging connects the vehicle to external power through EV supply equipment; onboard or external equipment manages conversion and control. Charging rate, connector/system compatibility, grid capacity, thermal limits, user behaviour and site design affect performance. Swapping exchanges standardised battery packs under controlled conditions.

Answer framework: Compare a charging scenario by power source, AC/DC conversion location, vehicle/EVSE communication, expected constraints, site data needed and safety controls—without calculating or instructing live installation.

Important point: Do not design, install or modify charging equipment from a handbook. Electrical installation and public charging require certified design, inspection, protection and regulatory compliance.

Explain Sustainability, lifecycle and safe response and state one correct method, application or precaution.−

EV sustainability depends on electricity source, material extraction, manufacturing, use efficiency, battery life, repair, second use and recycling. Safety planning covers normal operation, damaged vehicles, insulation faults, thermal events and emergency coordination under authorised procedures.

Answer framework: Create a lifecycle evidence map: stage, potential benefit/impact, data source, uncertainty, responsible stakeholder and improvement question. Add a clear escalation boundary for any high-voltage/battery abnormality.

Important point: If a battery is damaged, overheating, emitting odour/smoke or involved in a collision, follow the applicable emergency/manufacturer procedure and contact trained responders; do not improvise a repair or containment action.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: EV Architecture and Vehicle Performance.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Batteries, Packs, Thermal Design and BMS.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Motors, Controllers and Drivetrain Integration.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Charging, Infrastructure, Sustainability and Safety.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: EV types and architecture; key subsystems; traction/auxiliary energy paths; rolling, aerodynamic and grade resistance; power, torque and drive-cycle energy concepts..
2. Develop a complete answer or practical plan for: Cell and pack fundamentals; energy/power trade-off; battery parameters; SoC/SoH and self-discharge awareness; mechanical, electrical and thermal pack design; BMS functions..
3. Develop a complete answer or practical plan for: Electric-machine roles; torque, speed and back-EMF concepts; inverter/controller, DC/DC and onboard charger roles; regenerative braking; drivetrain and thermal integration..
4. Develop a complete answer or practical plan for: AC/DC charging, charging rate and infrastructure; swapping concepts; system communication/standardisation awareness; lifecycle/sustainability; inspection, incident awareness and safety boundaries..

LAST-MILE REVISION

Quick Revision

Module 1: EV Architecture and Vehicle Performance

EV types and architecture; key subsystems; traction/auxiliary energy paths; rolling, aerodynamic and grade resistance; power, torque and drive-cycle energy concepts.

- Match each topic to CO1.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Module 2: Batteries, Packs, Thermal Design and BMS

Cell and pack fundamentals; energy/power trade-off; battery parameters; SoC/SoH and self-discharge awareness; mechanical, electrical and thermal pack design; BMS functions.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Motors, Controllers and Drivetrain Integration

Electric-machine roles; torque, speed and back-EMF concepts; inverter/controller, DC/DC and onboard charger roles; regenerative braking; drivetrain and thermal integration.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Charging, Infrastructure, Sustainability and Safety

AC/DC charging, charging rate and infrastructure; swapping concepts; system communication/standardisation awareness; lifecycle/sustainability; inspection, incident awareness and safety boundaries.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
EV configurations and energy path	Battery-electric vehicles use a traction battery to supply an electric drive system.
Forces, torque, power and drive cycles	Vehicle energy demand is influenced by rolling resistance, aerodynamic drag, road grade, mass, acceleration and drive-cycle behaviour.
Battery parameters and pack architecture	A traction pack connects cells/modules with electrical protection, sensing, structural support, cooling/heating and high-voltage interfaces.
SoC, SoH, thermal management and BMS	State of charge estimates available energy; state of health indicates ageing/performance condition.
Traction motors and control	An electric traction motor converts electrical energy to wheel-driving mechanical energy.
Power-electronics and vehicle integration	Power electronics may include traction inverter, DC/DC converter and onboard charger.
Charging and swapping systems	Charging connects the vehicle to external power through EV supply equipment; onboard or external equipment manages conversion and control.
Sustainability, lifecycle and safe response	EV sustainability depends on electricity source, material extraction, manufacturing, use efficiency, battery life, repair, second use and recycling.

LEARNING RESOURCES

References

Recommended electric-vehicle references

1. I. Husain, Electric and Hybrid Vehicles: Design Fundamentals.
2. M. Ehsani et al., Modern Electric, Hybrid Electric, and Fuel Cell Vehicles.
3. J. Larminie and J. Lowry, Electric Vehicle Technology Explained.
4. Current manufacturer and regulatory service/safety documentation for any practical work.

Approved public electric-vehicle sources

- https://nptel.ac.in/courses/108106170
- https://afdc.energy.gov/vehicles/how-do-all-electric-cars-work

These sources provide the verified study basis while official Course 6023C detail is unavailable; they do not replace manufacturer repair information, high-voltage certification, emergency guidance or applicable electrical/transport regulations.